

Dice and dose-response relationships: infections

We have seen dose-response relationships in several contexts, from fertilizing with limiting nutrients increasing growth rates to increasing doses of toxins overwhelming an individual's capacity to tolerate it. Today we will think about dose-responses in pathogen or parasite infections¹.

Dice as pathogens

To develop a more intuitive sense of at least one way we can get a dose-response relationship, let's roll some dice. I like thinking about pathogens, so let's imagine that we are the hosts and each die is a single pathogen. A pathogen/die only causes an infection if it comes up a "6", otherwise you fight it off. We are going to simulate exposures to different doses of pathogens/dice to see how the probability of infection changes with the dose.

Of course a single host is either infected or uninfected, so a single person's data would either be a zero or a one (or a series of zeros and ones with repeated exposures). We want to determine what the *probability* of infection is at different doses, which we can estimate as the *proportion* of hosts² that are infected³ at each dose.

We will also consider two other scenarios: one where *two* sixes in a roll are needed to cause an infection and one where *four* are needed.

¹ The logic also holds for other cases where each challenge from something—a spark that might light a fire, a hit to the head that might cause a traumatic brain injury—has a small, independent chance of causing a catastrophe.

² Students rolling dice.

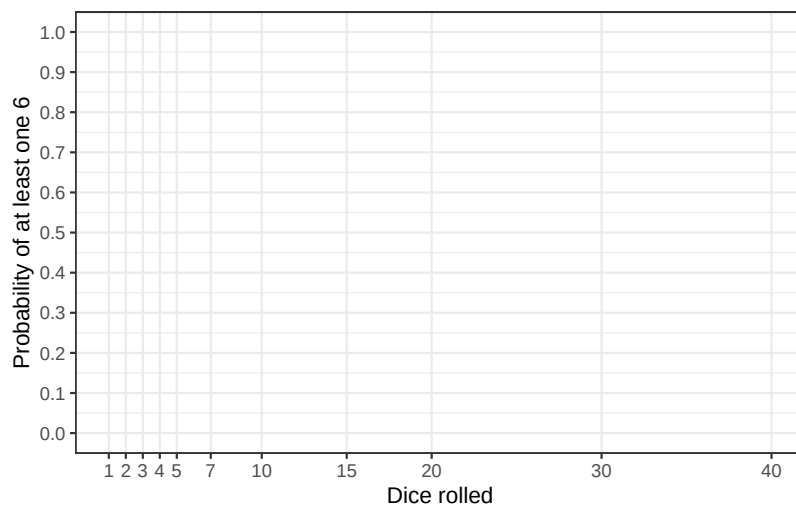
³ Roll at least one 6.

Before we start rolling our dice, I would like you to sketch out on the axes below your expectation for what these data are going to look like for each criterion for a successful infection. Be sure to think about what you will see for each of our criteria for success.

Number of students: _____

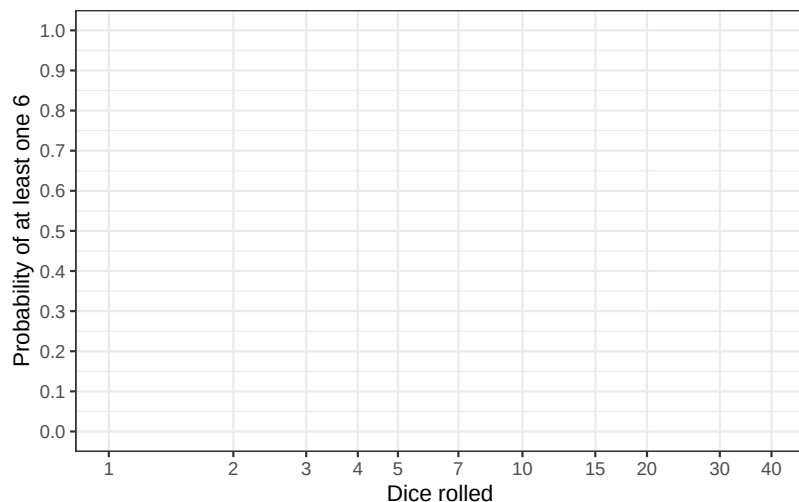
Dice rolled	Students with ≥ 1	Students with ≥ 2	Students with ≥ 4
1			
2			
3			
4			
5			
7			
10			
15			
20			
30			
40			

Then let's plot these data on the axes, below. (And yes, you will have to turn numbers of "infected" students into proportions of the class.)



Do your data look like what you expected? Why? What fits your expectation and what is a surprise?

Now plot the same data on the *logarithmic* x-axis, below. Note that distance along the x-axis now reflects *proportional* increases (e.g., doubling is always the same) rather than the increases in number.



Does this pattern fit your expectations better? Worse?

Useful and not

As always, our little toy model is useful in some ways and not-so-useful in other ways. Think about what was useful for helpful about this activity and what is missing, inaccurate, misleading, or otherwise not so helpful.